

Global Facilities – Chemical TMAH Standard

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1.0 Purpose

- 1.1 The objective of this document is to provide requirements on the Design, Operation, Maintenance, Delivery, Storage, Handling, Dispense, Distribution, Use, and Waste Collection of Tetramethylammonium Hydroxide (TMAH) in aqueous solutions or as a component of a mixture >1% concentration. This document covers the chemical dispense/blend/distribution systems, waste collection systems, exhaust systems, and process tools associated with TMAH, to mitigate safety and health hazards associated with potential exposure to TMAH.

2.0 Scope

- 2.1 This document covers all existing and new base build facilities system installations and also covers parts of process tool installation with regards to TMAH safety.
- 2.2 Site(s) impacted
 - 2.2.1 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites.

3.0 Definitions and Acronyms

Acronym	Definition / Description
AOV	Air Operated Valve
CCB	Clean Coupling Booth
CDU	Chemical Dispense Units
CoHE	Control of Hazardous Energy
ECTFE	Ethylene Chlorotrifluoroethylene (Halar)
EHS	Emergency, Health and Safety
EPDM	Ethylene Propylene Diene Monomer
EPR	Ethylene Propylene Rubber
FAT	Functional Acceptance Test
FCT	Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKM's
FEP	Fluorinated Ethylene Propylene (Teflon)
FMCS	Facility Monitoring and Control System
GFC	Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
GFTT	Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
Global Facilities Central Team	This is used interchangeably with Innovation & Integration Team

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Acronym	Definition / Description
Global Facilities Construction Team	This is used interchangeably with Global Construction Team
Global Facilities Operations Support Team	This is used interchangeably with Global Facilities Operations Team
Global Facilities Planning Team	This is used interchangeably with Global Facilities Line Planning Team
NC	Normally Closed
OAT	Operational Acceptance Test
PCTFE	Poly-Chloro-Trifluoro-Ethylene
PE	Polyethylene
PFA	Perfluoroalkoxy (Teflon)
PLC	Programmable Logic Controller
POC	Point of Connections
PP	Polypropylene
PTFE	Poly-Tetra Fluoro-Ethylene (Teflon)
PVC	Polyvinyl Chloride
SCADA	System Control and Data Acquisition
SSC	Stainless Steel Coated
TMAH	Tetra Methyl Ammonium Hydroxide
VMB	Valve Manifold Box

4.0 Roles and Responsibilities

Roles	Responsibility
Global Facilities Technology Team (GFTT)	- Define minimum technical requirements in the Global Facilities – Chemical-TMAH Standard and maintain the standard. - Participate in joint discussion and review pertaining to Global EHS - TMAH Safety requirements.
Global Environmental, Health and Safety (EHS)	- Communicates Global EHS - TMAH Safety Standard requirements to site EHS team and other corporate organizations (i.e., GFTT, PSM) - Provides direction, consultation, and guidance to Micron sites on program implementation - Participate in joint discussion and review pertaining to Global Facilities – Chemical- TMAH Standard. - Maintains Global EHS - TMAH Safety standard - Audits and inspects site's implementation of the program.
Global Process Safety Team (PSM)	- Participate in joint discussion and review pertaining to TMAH Safety requirements.
Site EHS, Environmental, Health and Safety	- Review all procedures related to tasks and systems where there is a risk of exposure to TMAH. - Review any physical changes to systems. - Review changes to procedures or tasks that may increase risk of exposure to TMAH. - Review any changes to chemical concentration present in a system. - Evaluate any necessary changes to Risk Assessments or Job Hazard Analyses.
Site System Engineer	- Develop step by step tasks for activities that have a potential for exposure to TMAH to mitigate risks identified in Risk Assessments or JHAs and to select appropriate PPE (with help from EHS if needed) - Review their system on a regular basis to ensure that it is proper working order and that there are no potential issues or gaps in maintenance procedures - Evaluate any necessary changes to Risk Assessments or Job Hazard Analyses

5.0 System Requirement Recommendations

5.1 Mechanical System Requirement

5.1.1 Installation of Exhaust for Facilities Systems and Process Tools

- The exhaust for any Facilities systems/equipment and process tools that utilize TMAH shall be connected to either Scrubbed Acid Exhaust or Scrubbed Caustic/Ammonia Exhaust. Scrubbed Caustic/Ammonia exhaust is the preferred system for connection to TMAH exhaust.
- TMAH exhaust ductwork materials of construction shall be Teflon-coated Stainless Steel (SSC).
- Install low point drain piping on all TMAH exhaust ductwork, to capture any potential condensation or liquid carryover.
- Low point drain piping materials of construction shall be Schedule 80 PVC. Exhaust drain piping does not require double containment.
- Avoid creating any low points or trapped sections when installing flexible ductwork at the process tool exhaust point of connections (POC). The lengths of the flexible ductwork installed shall be kept as minimum as possible. Transparent flexible ductwork material shall be considered (if available) to allow for visual inspection of possible condensation or liquid pooling. Sites shall have procedures established to safely drain out TMAH from flexible duct or when needed.
- Hazard warning labels shall be pasted on all Facilities system/equipment exhaust duct work, process tool exhaust ductwork, and exhaust drain piping that utilizes TMAH. For more details, refer to **section 5.5.3** under "Labels for Waste Stream and Exhaust Ductwork/Condensate Drain".

5.2 Instrumentations & Control System Requirement

5.2.1 Design Parameters

- a) All TMAH or TMAH as a component of a mixture >1% concentration - Chemical Dispense Units (CDUs), Blend Units, Valve manifold boxes (VMBs), and other containment boxes that contain mechanical joints shall have chemical compatible leak detection device installed.
- b) All leak detection devices shall be connected to and monitored at the site chemical system SCADA (System Control And Data Acquisition).
- c) Leak detection shall be wired in a normally closed (NC) condition so that the circuit is supervised.

5.2.2 Operating Parameters

- Leak detection shall be maintained in good working order and tested periodically or during preventive maintenance based on site-specific safety requirements.
- System response to leak detect alarms shall be determined by site EHS and Chemical Services policies and any applicable national, international, or local safety codes and requirements.

5.2.3 Design Configuration

- Distribution systems shall utilize point of use float switches or alternative devices in low point drains, junction boxes or VMBs.

5.2.4 QA/Testing

- Operational Acceptance Tests (OAT)/Functional Acceptance Tests (FAT) shall be performed on leak detection devices and systems prior to chemical commissioning and verified that devices/systems respond in the manner intended by design. Testing shall include physical activation from the device/system to the chemical dispense/Blend system (if applicable) and to the SCADA system.
- Based on local code or regulations and/or site requirements, periodic testing of leak detect devices may be required. Testing shall not impact production unless required by said regulations.

5.3 Chemical System Requirement

5.3.1 Design Parameters

- All TMAH or TMAH as a component of a mixture >1% concentration - dispense, blend and distribution systems shall be designed according to IBC, IFC, NEC, NFPA, and other applicable local codes and regulations and adopted equivalent standards.

5.3.2 Operating Parameters

- The **delivery pressure** shall be controlled to avoid exceeding the maximum rated pressure of any component installed within any Facilities system and Process tool that utilizes TMAH in aqueous solutions or as a component of a mixture >1% concentration.

5.3.3 Operation

- **Drum and Totes Connection**
 - Spray/splash guards shall be installed around the drum or tote connections to protect personnel in the area. Technicians making drum and tote connections shall wear the Minimum Corporate Required PPE as stipulated in Global EHS - TMAH Safety Standard.
- **Tanker Truck Unloading:**
 - Offload pressure to the tanker truck or ISO container shall be set at a value not to exceed the manufacturer's recommended pressure.
 - Spray/splash guards shall be installed around the flange connections in the offload box to protect personnel in the area. If quick connectors are used, no spray/splash guards are needed however, shall wear the Minimum Corporate Required PPE as stipulated in [Global EHS - TMAH Safety Standard](#).
- **Sample Collection:**
 - Technicians performing sample collection shall wear the Minimum Corporate Required PPE stipulated in [Global EHS - TMAH Safety Standard](#).

5.3.4 Maintenance

- TMAH as a component of a mixture >1% concentration - dispense, and distribution systems, including valves and other main components, shall be maintained based on the O&M manual and manufacturer's O&M manuals.
- Maintenance procedures shall include, but not be limited to the following items:
 - Technicians performing maintenance shall wear the Minimum Corporate Required PPE as stipulated in [Global EHS - TMAH Safety Standard](#).
 - Chemical Dispense Units (CDUs), Blend and Distribution systems shall be depressurized before any wet component replacement, or if not possible, wet component change shall be carried out where an appropriate manual isolation valve is available while wearing appropriate
 - COHE (Control of Hazardous Energy) procedures for expansion and/or maintenance as per established site practice.
 - Isolation of one part of the distribution line for maintenance and/or expansion, where an isolation valve is available.
 - Inspection of mechanical joints, valves, and other main components as well as secondary containment shall be performed during routine maintenance to prevent major failure.
 - An emergency recovery procedure shall be implemented to guarantee a safe and efficient response to a TMAH Chemical leak/spill event. It shall be coordinated using the TMAH Emergency Response / Spill and TMAH Emergency Response/First Aid - Medical reference sheet as stipulated in [Global EHS - TMAH Safety Standard](#).

5.3.5 Material Compatibility

- TMAH as a component of a mixture is incompatible with materials as listed below. All wetted components as well as non-wetted components that has a potential for external exposure (e.g. pump external PVDF nuts and valve external PVDF union nut, etc.) shall avoid incompatible materials as this may Cause premature failure, increasing hazard.
- The list of material compatibility for aqueous solutions of TMAH indicated below is based on data from SACHEMS that has experience in the manufacture/handling of TMAH. Refer to **Table 1**.

Table 1: Material compatibility list for aqueous solutions of TMAH

Compatible	Incompatible
a) Non-Metal: ○ PP (High Purity grade only) ○ PE (High Purity grade only) ○ PVC ○ PTFE ○ PFA ○ ECTFE (Halar) ○ PCTFE ○ FEP b) Elastomers: ○ EPDM ○ EPR ○ Kalrez/ Chemraz c) Metal: ○ 304/316 Stainless Steel ○ Nickel 200 ○ Hastelloy-type alloys	a) Non-Metal: ○ PVDF b) Elastomers: ○ Buna-N ○ Viton ○ Silicon c) Metal: ○ Aluminium ○ Copper ○ Carbon Steel

5.4 Chemical Installation Requirement

5.4.1 Dispense/Blend Unit

- Dispense/Blend units shall be provided with door interlocks and leak detection for all enclosures where mechanical connections and single contained tubing are present. Door interlock and leak detection shall immediately close automated chemical supply valves, initiate a local alarm, shut down supply and de-pressurize system. If systems are designed with isolated primary and secondary engines for the purpose of

redundancy with separate engine leak detection, the affected engine shall de-pressurize and shutdown without impacting the redundant engine.

- Removable spray/splash guards shall be used to shield technicians from pressurized components during maintenance.
- Dispense/Blend units shall be equipped with sampling devices designed to minimize contamination and exposure during sample collection.

5.4.2 Day Tank

- Spray or splash guards shall be installed on any connections on the top or bottom of the day tanks. The guards can be heavy vinyl curtains, white/clear PVC barriers, or flange/valve safety spray shields. All transfer/recirculation/inter-connect PFA (primary) lines shall not be exposed and shall have unbroken continuous double containment up to all the connection points within the spray/splash guards.
- External sight tubes shall also require spray/splash protection. The preferred method is to have a clear PVC double containment for the sight tubes (schedule 40 or greater).

5.4.3 Drum/Tote Stations

- Spray/splash guards shall be installed around the drum or tote connections or shall have a shielding to enclose the drum/tote. The shielding can be heavy vinyl curtains or poly/clear PVC barriers around the drum/tote.
- The drum/tote PFA (primary) transfer/recirculation lines shall not be exposed and shall have an unbroken continuous double containment up to all the connections points within the spray/splash guards.

5.4.4 Distribution System

- Distribution systems shall be individually contained and not allowed to be routed in chemical trays. Existing distribution, PFA (primary) lines shall not be exposed and shall be re-routed with individual double containment in a continuous unbroken clear PVC secondary containment (schedule 40 or greater) from the CDU to each valve manifold box.
- Valve manifold boxes (VMB) doors must be secured with mechanical latches.
- Valve manifold boxes (VMB) shall have leak detection in place.
 - If VMB is designed with a single leak sensor, the CDU shall de-pressurize and stop supply upon leak activation. The VMB shall have appropriate manual isolation valves in place to safely replace the leaking valves or rectify leaking fittings with PPE.
 - If VMB is designed with dual leak detection system, it shall be configured as low level (warning) and high level (shutdown). Upon activation of both low and high level leak sensors, the CDU shall de-pressurize and stop supply. The VMB shall have appropriate manual isolation valves in place to safely replace the leaking valves or rectify leaking fittings with PPE.
 - If VMB is designed with dual leak detection system with an automatic VMB isolation valve, upon activation of low-level leak sensor, the automatic VMB isolation valve shall close to stop supply to VMB. When both low and high level leak sensors are activated, the CDU shall de-pressurize and stop supply. The VMB shall have appropriate manual isolation valves in place to safely replace the leaking valves or rectify leaking fittings with PPE.
- All instruments or other mechanical connections shall be installed with secondary containment and/or containment boxes.

5.4.5 Others

- Delivery
 - Drums and totes shall be delivered to the site with connections covered.
 - All totes/drums shall be delivered with dip tubes pre-installed.
 - Bottom drain valves of totes shall be protected during transport. If possible, it is preferred not to have a manual drain valve at the bottom of totes to eliminate the risk of leaks.
 - Totes shall be delivered in a protective cage where possible, to minimize puncture risk from forklifts.
 - Refer to [Global EHS - TMAH Safety Standard](#) for more details on safe operation during delivery.

- Storage
 - Separate TMAH from other incompatible chemicals by distance or physical separation as required by building/local codes.
 - Storage areas shall be well maintained. Adequate space shall be provided for the equipment used to transport chemicals safely.
 - Containers shall be tightly closed. TMAH readily absorbs carbon dioxide from the air. Loose caps present a spill risk.
 - TMAH shall be stored on the floor or bottom rack of shelving units.
 - If stored on higher shelves, the racking system shall be of an open grating style.
- System Exhaust
 - System exhaust from TMAH CDU/tote/drum enclosures shall be connected to scrubbed caustic/ammonia exhaust.
 - All systems shall have leak detection as specified in [UHP Chemical Distribution Best Known Methods \(BKM\) Guidelines](#)

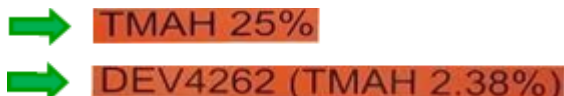
5.5 Chemical Systems Labelling Requirement

5.5.1 TMAH labels and Hazard warning labels must clearly identify areas where TMAH or TMAH as a component of a mixture with concentrations >1% may be encountered in facilities, sub-fab and production areas. Sites, if required, may use both local and English language for TMAH Hazard warning labels. Both languages may be incorporated in the same label space or as a separate label, if constrained by size and the main requirement is to ensure that the label is clearly visible and legible from ground level.

5.5.2 Labels for Facilities Chemical System and Process Tools (located in Fab & Sub-fab):

- TMAH labels shall clearly indicate both the Chemical name/notation and TMAH % concentration in orange label with black text (See **Figure 1** for an example of a label for TMAH 25% or DEV4262 (TMAH 2.38%))

Figure 1: Example of Labeling convention



- The labels shall be clearly legible and visible and shall be pasted on all of the following systems
 - Bulk Storage/ Day tanks
 - Chemical Dispense Unit
 - Blend Unit
 - Drum/Tote Transfer Unit/Stations

- Clean Coupling Booth (CCB)
- VMB, Tee/Junction box, End/Expansion-box for distribution and Containment boxes at top and bottom of storage/Blend tank/Day tank.
- Double contained supply/return piping and all other system inter-connect contained piping including hook-up lines from VMB to tool chemical cabinet and/or process tool.
- Process Tool chemical cabinet and Main Process tools.
- All double contained piping shall be labeled with the chemical contained and the direction of flow. Two sizes of labels are available: for 1-inch secondary containment and containment that is greater than 1 inch.
 - Piping = 1-inch, the lettering shall be bold, black, non-serif, 1-inch high (78 pt), Helvetica font on a 1.5-inch wide orange adhesive label.
 - Piping >1 inch, the lettering shall be bold, black, non-serif, 1.5-inches high (114 pt), Helvetica font on a 2.5-inch wide orange adhesive label.
- All containments shall be labeled at every 5 meters or shorter and on both sides of any penetrations or changes in direction. Labels shall be clearly visible and legible at ground level.
- As a minimum guideline, all other systems indicated above shall be labeled with bold, black, non-serif, 1 inch high (78 pt), Helvetica font on a 1.5-inch wide orange adhesive label.
- The orange hazard warning label to indicate the risk of TMAH exposure shall be placed on all equipment access panels/doors on all of the above-mentioned systems with the exception of double contained piping. The hazard warning label shall have the verbiage "WARNING - CONTAINS TMAH - AVOID SKIN CONTACT" in orange label and black text. See Figure 2. The label shall be sized such that is clearly visible and legible from ground level and shall be placed in close proximity to equipment access panel/door's handle/knob where possible or in clear sight of the operator.

Figure 2: Example of Labeling convention

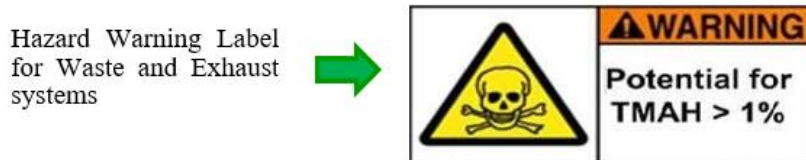


- Black and orange tape (zebra tape) as shown above shall be placed on the floor/tiles around the perimeter of Process tool chemical cabinets and Main Process tools or around the specific section of the system/tool that contains TMAH or TMAH as a component of a mixture with concentrations >1%.

5.5.3 Labels for Waste Stream and Exhaust Ductwork/Condensate Drain.

- Hazard warning labels shall be placed on Exhaust ductwork at tool POC, Exhaust condensate drains and Waste streams to identify any exhaust ductwork and drain lines that has a "Potential" for TMAH concentrations >1%. See **Figure 3**

Figure 3: Labels for Waste Stream and Exhaust Ductwork/Condensate Drain



- The hazard warning label indicating "Potential for TMAH concentrations >1%" shall be used for the following:
 - Process tool drain
 - Tool chemical cabinet/support equipment drain lines
 - Drain lines from Chemical dispense unit (CDU), Blend unit, Drum/Tote Transfer unit/station
 - Room sump pump and drain lines
 - Valve Manifold Box (VMB) drain line (if connected to submain)
 - Submain and hook-up drain lines
 - Final collection/storage, treatment tanks (if applicable to site)
 - Point of connection (POC) exhaust ductwork at Process Tool
 - Point of connection (POC) exhaust ductwork at Chemical Distribution System
- For drain pipes /POC exhaust duct size that are greater than 3 inch and for collection tanks: The total dimension of the above label shall be 250mm by 100mm and it shall include the signal word for warning, identification of the hazard, and a pictogram to display the consequence.
 - Signal word "WARNING" shall be in black lettering of 3/4 inch high, bold in Arial font on top right-hand corner with an orange background of dimension 140mm x 35mm.
 - Hazard Identification (spelling) "Potential for TMAH >1%" shall be in black lettering of 5/8 inch high, bold in Arial font at bottom right corner with a white background of dimension 140mm x 65mm.
 - Pictogram indicating toxicity on the left with dimension of 110mm x 100mm.

- For drain pipes/POC exhaust duct size below 3 inch, the total dimension of the above label shall be 125mm by 50mm and the fonts sized to match the label dimension.
- For smaller diameter drain pipes where the label size is too big, it shall be laminated with labels at both front and back and cable tied to the pipe.
- All submain drain lines shall be labeled at no more than 20 ft and on both sides of any penetrations or changes in direction. Labels shall be clearly visible and legible at ground level.
- Tool hook-up drain line shall be labeled at both tool POC and at the connection to submain (Y-Tee or Tee connection).
- All hazard warning labels shall be clearly visible and legible from ground level.

5.5.4 TMAH Systems Labelling Examples

Refer to the figures below for sample of acceptable TMAH Labeling for Facilities Systems & Process Tools across the Micron Sites.

Figure 4a: Sample of acceptable TMAH Labeling for Facilities Systems & Process Tools across the Micron Sites.

TMAH Labels on Hook-up line

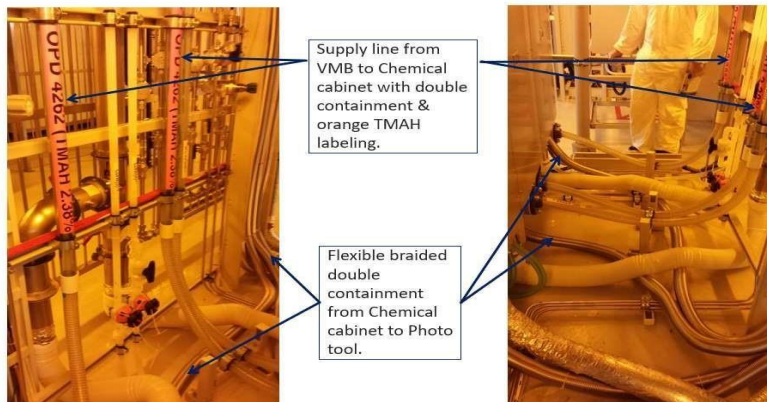


Figure 4b: Sample of acceptable TMAH Labeling for Facilities Systems & Process Tools across the Micron Sites.

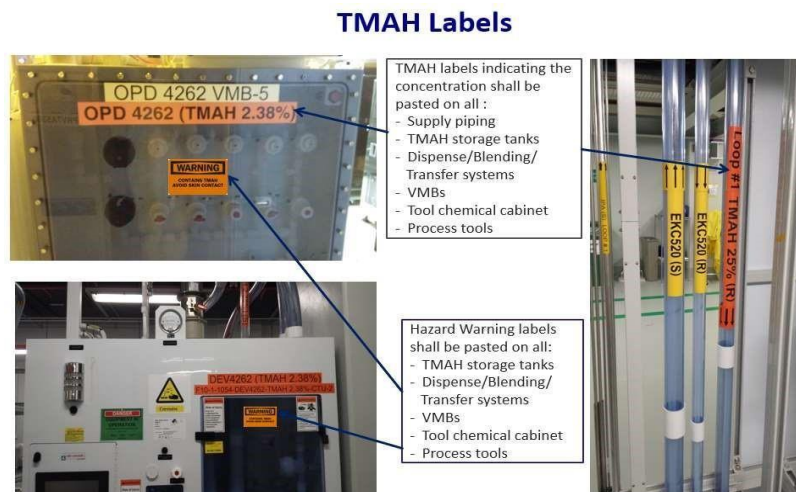


Figure 4c: Sample of acceptable TMAH Labeling for Facilities Systems & Process Tools across the Micron Sites.



Figure 4d: Sample of acceptable TMAH Labeling for Facilities Systems & Process Tools across the Micron Sites.

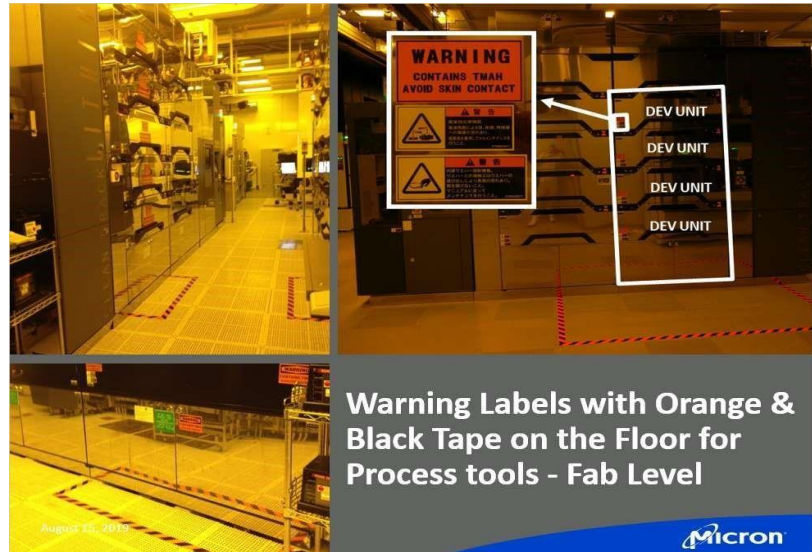


Figure 4e: Sample of acceptable TMAH Labeling for Facilities Systems & Process Tools across the Micron Sites.

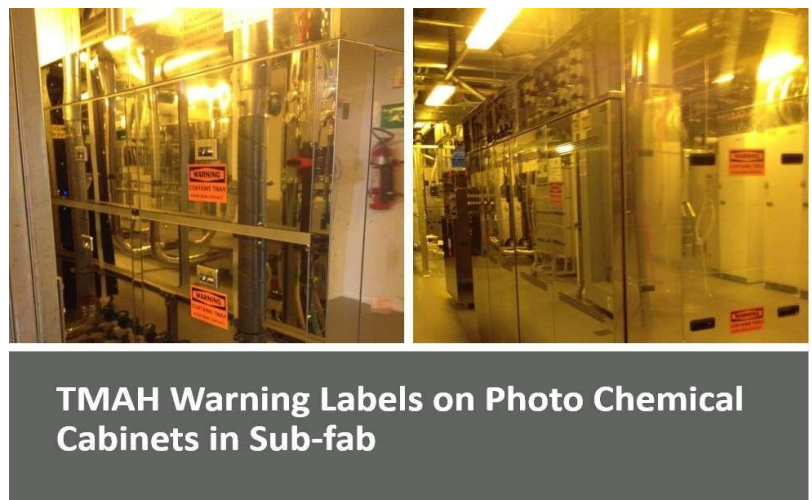


Figure 4f: Sample of acceptable TMAH Labeling for Facilities Systems & Process Tools across the Micron Sites.



5.6 Water & Waste Water

5.6.1 Maintenance

- Maintenance, repair, and construction of drains shall include, but not be limited to, the following items:
- Technicians performing maintenance, repair, and construction shall wear the Minimum Corporate Required PPE as stipulated in [Global EHS - TMAH Safety Standard](#).
- Isolation or diversion valves shall be installed on the main drain line downstream of the last sub main or lateral connection and prior to the at end of the drain lines above the collection tanks to allow for maintenance.
- Maintenance, repair and construction on a drain lateral shall require isolation of chemical supply to the tools connected to the drain lateral. Chemical supply shall be isolated in accordance with site CoHE (Control of Hazardous Energy) procedures prior to work on a drain lateral.
- An emergency recovery procedure shall be implemented to guarantee a safe and efficient response to a TMAH chemical leak/spill event. It shall be coordinated using the TMAH Emergency Response/ Spill and TMAH Emergency Response/First Aid - Medical reference sheet as stipulated in [Global EHS - TMAH Safety Standard](#).

5.6.2 Materials

- PVDF material is not suitable for TMAH waste lines. For more details, refer to **Section 5.3.5** on "Material Compatibility".

5.6.3 Installation

- Labels:
 - All drain lines for Manufacturing tools or associated tool support equipment supplied by any combination of chemistries that include TMAH at concentrations above 1% shall be labeled with "Potential for TMAH >1%" hazard warning label and the direction of flow.
 - TMAH waste collection drains shall be identified with "Potential for TMAH >1%" hazard warning label and the direction of flow from the point of connection at the tool to the eventual point of final collection, storage, or treatment to eliminate the hazards related to this chemistry. However, if the TMAH concentration is deemed to be < 1% at the final Wastewater Treatment, no labeling is required.
 - The waste streams/systems shall be labeled but not limited to the following:
 - Process tool drain lines.
 - Tool chemical cabinet/support equipment drain lines
 - Drain lines from Chemical dispense unit (CDU), Blend unit, Drum/Tote Transfer unit/station.
 - Room sump pump and drain lines.
 - Valve Manifold Box (VMB) drain line (if connected to submain).
 - Submain and hook-up drain lines.
 - TMAH waste collection tanks and associated lines (applicable to sites that has a dedicated waste TMAH collection and treatment system).
 - For more details of labeling guidelines, refer to **Section 5.5** under "Chemical Systems Labelling Requirements".
- Installation of Waste TMAH Lines
 - Gravity drain lines carrying TMAH waste shall be hard piped connections to the facility drain system. The piping material shall be PVC schedule 80 or HI-PVC or PP.
 - Pressurized drain/transfer lines that utilizes PFA tubing, PVC or PP to contain the waste stream, shall have either PE (for outdoor use) or clear PVC (for indoor use) as a double containment. The double containment shall be unbroken continuous containment up to all the connection points.
 - All pressurized drain/transfer lines that are hard piped without double containment shall have a splash/spray guard around the flange connections.

5.7 Process Tools

5.7.1 Process Tool - Design Configuration

All Process tools using TMAH or TMAH as a component of a mixture >1% concentration:

- All process tools shall protect and contain all process piping within the tool or chemical cabinet enclosure.
- The tool and chemical cabinets shall be provided with door interlocks for hinged door with latch/ handle and leak detection for all enclosures where mechanical connections and single contained tubing are present. Door interlock and leak detection shall immediately close automated chemical supply valves on the tool, initiate a local, alarm, notify operator/owner and shut down pneumatic supply to the tool chemical pumps and/or de-pressurize tool pressure pot systems.

5.7.2 Process Tools - Maintenance

- TMAH Process tools shall be maintained per manufacturer's recommendations.
- Maintenance procedures shall include, but not be limited to the following items:
 - Technicians performing maintenance shall wear the Minimum Required PPE as stipulated in [Global EHS - TMAH Safety Standard](#).
 - Areas where chemicals containing TMAH may be pressurized, it must be depressurized and LOTO applied prior to maintenance work. CoHE procedures for maintenance must be in place.
 - Means shall be available to verify all components have been depressurized before removing any protective cover.
 - Where necessary to read gauges or displays behind covers, covers shall be clear and adequately transparent to read the gauge or display without removing the cover.
 - Components that may need to be dismantled for maintenance, or repair, or replacement shall be flushed prior to being dismantled.
 - Components or fittings that need to be disconnected during maintenance shall be drained and will not result in liquid spilling onto maintenance personnel.
 - An emergency recovery procedure shall be implemented to guarantee a safe and efficient response to a TMAH chemical leak/spill event. It shall be coordinated using the TMAH
 - Emergency Response/Spill and TMAH Emergency Response/First Aid - Medical reference sheet as stipulated in [Global EHS - TMAH Safety Standard](#).

5.7.3 Process Tools - Installation

- All process piping running to the process tools or chemical cabinets and tool inter-connects shall be installed with continuous secondary containment capable of directing liquid to a cabinet or enclosure with leak detection.
- Drain lines carrying TMAH waste shall be hard piped and/or double contained to the facility drain system. Refer to **Sections 5.6.2 and 5.6.3** for more details
- Any drip pans installed under the tools shall have a hard piped and/or double contained connection to the facility drain system.
- A leak detection sensor shall be installed in the drip pan. There shall be no mechanical fittings outside of containment. All connections through containment pans must be welded feed-through type connections.
- Labels highlighting the presence of TMAH shall be pasted on process tools, chemical cabinets, all double contained inter-connect piping and drain lines. Refer to **Section 5.5** on "Chemical Systems Labelling Requirements" for more details on labelling guidelines.
- Secondary containment for tool hook-up lines and tool chemical cabinets in the sub-fab shall utilize schedule 40, clear PVC and/or shall utilize Poly Vinyl breasted flexible type if constrained by space and installation difficulties. The secondary containment connections shall be secured by utilizing appropriate bulk head connections where possible or pipe clips.
- Process tool exhaust ductwork and exhaust drain piping utilizing TMAH shall be labeled in accordance with **Section 5.5**. Exhaust condensate drain piping does not require double containment or drip pan.
- For more details of labeling guidelines, refer to **Section 5.5** on "Chemical Systems Labelling Requirements" for more details on labelling guidelines.

6.0 Best Practices - applicable for new system installations if it does not exist in existing systems and over and above the minimum requirements already described.

6.1 Chemical - Operation

6.1.1 Drum and Totes Connection:

- Quick disconnects shall be used for dip tube connection.

6.1.2 Tanker Truck Unloading:

- Quick connect coupler connections shall be used for unloading to Clean Coupling Booths (CCBs).
- Safety clamps shall be used to prevent the unloading hose with coupler from dislodging from the CCB (high pressurized hoses could endanger personnel if dislodged during high pressure unloading operations).

6.2 Chemical - Installation

6.2.1 Drums/Totes Stations:

- Drums or totes of TMAH or TMAH as a component of a mixture with a concentration >1% will be installed in exhausted enclosures with leak detection. The enclosure shall have a built-in double glove access port with chemical resistant gloves attached.

6.2.2 TMAH Chemical Dispense /Blend Units:

- Chemical dispense/blend units shall be installed in chemical compatible, exhausted enclosures.
- The enclosures shall have leak detection sensors to stop chemical supply upon activation.
- The enclosures shall be compartmentalized to segregate between primary and secondary engines with a permanent spray/splash guards or a separation plate to shield technicians from pressurized components during maintenance.
- Chemical dispense/blend units shall be equipped with sampling compartment with door interlocks designed to stop sample collection when sample door is opened. This is to minimize TMAH exposure during sample collection.

6.2.3 Day Tanks:

- A permanent enclosure with leak detection sensors shall be installed around the top and bottom connections of the day tanks.

6.2.4 Distribution Systems:

- For all new installations, TMAH distribution PFA piping shall be double contained in a continuous PVC secondary containment (schedule 40 or greater) and all POC valves, instruments, or other mechanical connections are to be enclosed in a valve manifold box (VMB) or Tee-box with leak detection to stop the chemical supply upon activation.
- Each POC stick from the main manifold inside the VMB shall comprise of a manual valve (MV) followed by a sample/purge tee-off valve and finally an air operated valve (AOV). The AOV shall operate based on tool demand.
- VMB access cover panels shall have keyed locks in place to prevent unauthorized access.

6.3 TMAH Standard Checklist (check for compliance)

6.3.1 A TMAH Standard checklist can be found under **Appendix** to check for compliance and can be used as guideline for audit.

7.0 Related Documents

- 7.1 [Materials of Construction \(NON-High Purity Applications\) SACHEM](#)
- 7.2 [Global EHS - TMAH Safety Standard](#)
- 7.3 [Waste Water Reuse/Reclaim Best Known Methods Guideline](#)
- 7.4 [UHP Chemical Distribution Best Known Methods \(BKM\) Guidelines](#)
- 7.5 [AE / AME Best Known Methods \(BKM\) Guidelines](#)

8.0 Document Control

8.1 **Approval:** GLOBAL_FAC_GFTT_DOC_APPR

8.2 **Notification:**

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_EHS_TEAM_MEMBERS
MCT_CPEE_PHOTO
PSM, KEG_WP_ENG
MCT_PEE_MATENG
FCG_F2_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10W_FAC_NOTIFY
FCG_F10N_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MTTW_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY
MTB_FAC_ALL_NOTIFY

8.3 **Final Viewer Access**

All Micron Team Members at all Sites

8.4 **Retention**

Adherence to the requirements specified in the
[Global Facilities – Document Control Procedures](#)

8.5 **Review**

Biennially (two years)

9.0 Revision History

Rev #	Section Changed	Details of Changes	Date
0	NA	New document	07/09/2012
1	Sections 1.3, 2.1.1, 2.4.10, 2.4.12 (Distribution system-1), 2.7.3 (2), 2.7.12 (2,5,6), 3.4.12 (Distribution system-1,2) and 6.1 Sections 2.6.8, 2.6.9, 2.6.12, 2.6.13 added	Revision done on the sections. New sections added	07/09/2013
2	Sections 1.3, 2.1.1, 2.1.2, 2.1.12, 2.4.1, 2.4.7, 2.4.8, 2.4.9, 2.4.12, 2.4.14, 2.6.8, 2.6.12, 2.7.3, 2.7.12, 3.4.12 Sections 2.1.12, 2.6.9, 2.7.8, 2.7.14 Section 2.6.13 Section 3.4.14 Section 2.1.1 Section 2.1.2	Revision done on the sections. New sections added Section removed Removed and transferred to section 2.4.14 Removed and transferred to section 1.3 Removed and transferred to section 2.4.2	8/26/2019
3	Multiple sections of the document	Document name change from TMAH FOR Standard to TMAH BKM Standard Upgrade document to a controlled document through ECN system.	04/13/20
4	Main page Multiple sections of the document Section 8	Document name change from TMAH BKM Standard to Global Facilities – Chemical - TMAH Standard Document format changed to System Standard format Update notification group to include MCT, PSM and EHS Teams	04/23/20

APPENDIX

Global Facilities - TMAH Standard Checklist

Item	Description	This section can be used as a check for compliance to Global Facilities –TMAH Standard for new/existing Facilities systems & Process Tools	
		Compliance Status (YES/ NO/ NA)	Comment (If No, provide rectification plan)
1. General Information			
1.1 Description	The objective of this document is to provide requirements on the Design, Operation, Maintenance, Delivery, Storage, Handling, Dispense, Distribution, Use, and Waste Collection of Tetramethylammonium Hydroxide (TMAH) in aqueous solutions or as a component of a mixture >1% concentration. This document covers the chemical dispense/blend/distribution systems, waste collection systems, exhaust systems, and process tools associated with TMAH, to mitigate safety and health hazards associated with potential exposure to TMAH.		
2. Minimum			
2.1 Mechanical			
2.1.12 Installation	<u>Exhaust for Facilities Systems and Process Tools:</u> 1. The exhaust for any Facilities systems/equipment and process tools that utilize TMAH shall be connected to either Scrubbed Acid Exhaust or Scrubbed Caustic/Ammonia Exhaust. Scrubbed Caustic/Ammonia exhaust is the preferred system for connection to TMAH exhaust. 2. TMAH exhaust ductwork materials of construction shall be teflon-coated Stainless Steel (SSC). 3. Install low point drain piping on all TMAH exhaust ductwork, to capture any potential condensation or liquid carryover. 4. Low point drain piping materials of construction shall be Schedule 80 PVC. Exhaust drain piping does not require double containment. 5. Avoid creating any low points or trapped sections when installing flexible ductwork at the process tool exhaust point of connections (POC). The lengths of the flexible ductwork installed shall be kept as minimum as possible. Transparent flexible ductwork material shall be considered(if available) to allow for visual inspection of possible condensation or liquid pooling. Sites shall have procedures established to safely drain out TMAH from flexible ductwork when needed. 6. Hazard warning labels shall be pasted on all Facilities system/equipment exhaust ductwork, process tool exhaust ductwork, and exhaust drain piping that utilizes TMAH. For more details, refer to section 2.4.12 under "B) Labels – Waste Stream and Exhaust Ductwork/Condensate Drain".		

2.2 Electrical			
2.3 I & C – Facilities Chemical System			
2.3.1 Design Parameters	1. All TMAH or TMAH as a component of a mixture >1% concentration – Chemical Dispense Units (CDUs), Blend Units, Valve manifold boxes (VMBs), and other containment boxes that contain mechanical joints shall have chemical compatible leak detection device installed. 2. All leak detection devices shall be connected to and monitored at the site chemical system SCADA (System Control And Data Acquisition). 3. Leak detection shall be wired in a normally closed (NC) condition so that the circuit is supervised.		
2.3.2 Operating Parameters	1. Leak detection shall be maintained in good working order and tested periodically or during preventive maintenance based on site-specific safety requirements. 2. System response to leak detect alarms shall be determined by site EHS and Chemical Services policies and any applicable national, international, or local safety codes and requirements.		
2.3.3 Design Configuration	1. Distribution systems shall utilize point of use float switches or alternative devices in low point drains, junction boxes or VMBs.		
2.3.4 QA/Testing	1. Operational Acceptance Tests (OAT) / Functional Acceptance Tests (FAT) shall be performed on leak detection devices and systems prior to chemical commissioning and verified that devices/ systems respond in the manner intended by design. Testing shall include physical activation from the device/system to the chemical dispense/Blend system (if applicable) and to the SCADA system. 2. Based on local code or regulations and/or site requirements, periodic testing of leak detect devices may be required. Testing shall not impact production unless required by said regulations.		

2.4 Chemical			
2.4.1 Design Parameters	1. All TMAH or TMAH as a component of a mixture >1% concentration – dispense, blend and distribution systems shall be designed according to IBC, IFC, NEC, NFPA, and other applicable local codes and regulations and adopted equivalent standards.		
2.4.2 Operating Parameters	1. The delivery pressure shall be controlled to avoid exceeding the maximum rated pressure of any component installed within any Facilities system and Process tool that utilizes TMAH in aqueous solutions or as a component of a mixture >1% concentration.		
2.4.3 Operation	<u>Drum and Totes Connection:</u> 1. Spray/splash guards shall be installed around the drum or tote connections to protect personnel in the area. Technicians making drum and tote connections shall wear the Minimum Corporate Required PPE as stipulated in Global EHS – TMAH Safety Standard <u>Tanker Truck Unloading:</u> 1. Offload pressure to the tanker truck or ISO container shall be set at a value not to exceed the manufacturer's recommended pressure. 2. Spray/splash guards shall be installed around the flange connections in the offload box to protect personnel in the area. If quick connectors are used, no spray/splash guards are needed however, shall wear the Minimum Corporate Required PPE as stipulated in Global EHS – TMAH Safety Standard <u>Sample Collection:</u> 1. Technicians performing sample collection shall wear the Minimum Corporate Required PPE as stipulated in Global EHS – TMAH Safety Standard		
2.4.4 Maintenance	1. TMAH or TMAH as a component of a mixture >1% concentration – dispense, blend and distribution systems, including valves and other main components, shall be maintained based on the O&M manual and manufacturer's O&M manuals. 2. Maintenance procedures shall include, but not be limited to the following items: 2.1 Technicians performing maintenance shall wear the Minimum Corporate Required PPE as stipulated in Global EHS – TMAH Safety Standard. 2.2 Chemical Dispense Units (CDUs), Blend and Distribution systems shall be depressurized before any wet component replacement, or if not possible, wet component change shall be carried out where an appropriate manual isolation valve is available while wearing appropriate PPE. 2.3 CoHE(Control of Hazardous Energy) procedures for expansion and/or maintenance as per established site practice. 2.4 Isolation of one part of the distribution line for maintenance and/or expansion, where an isolation valve is available. 2.5 Inspection of mechanical joints, valves, and other main components as well as secondary containment shall be performed during routine maintenance to prevent major failure. 2.6 An emergency recovery procedure shall be implemented to guarantee a safe and efficient response to a TMAH chemical leak/spill event. It shall be coordinated using the TMAH Emergency Response / Spill and TMAH Emergency Response/First Aid – Medical reference sheet as stipulated in Global EHS –TMAH Safety Standard.		

2.4.5 Materials

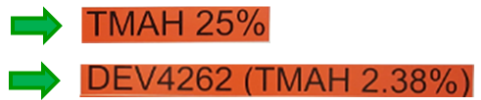
Material Compatibility:

1. TMAH or TMAH as a component of a mixture is incompatible with materials as listed below. All wetted components as well as non-wetted components that has a potential for external exposure (e.g. pump external PVDF nuts and valve external PVDF union nut, etc.) shall avoid incompatible materials as this may cause premature failure, increasing hazard.

2. The list of material compatibility for aqueous solutions of TMAH indicated below is based on data from SACHEMS that has experience in the manufacture/handling of TMAH.

COMPATIBLE	INCOMPATIBLE
a) Non-Metal: PP, Polypropylene (High Purity grade only) PE, Polyethylene (High Purity grade only) PVC, Polyvinyl Chloride PTFE, Poly-Tetra Fluoro-Ethylene (Teflon) PFA, Perfluoroalkoxy (Teflon) ECTFE, Ethylene Chlorotrifluoroethylene (Halar) PCTFE, Poly-Chloro-Trifluoro-Ethylene FEP, Fluorinated Ethylene Propylene (Teflon)	a) Non-Metal: PVDF, Polyvinylidene Fluoride (Kynar)
b) Elastomers: EPDM, Ethylene Propylene Diene Monomer EPR, Ethylene Propylene Rubber Kalrez/Chemraz	b) Elastomers: Buna-N Viton Silicon
c) Metal: 304/316 Stainless Steel Nickel 200 Hastelloy-type alloys	c) Metal: Aluminium Copper Carbon Steel

2.4.6 Installation	<u>Dispense/Blend Unit:</u> 1. Dispense/Blend units shall be provided with door interlocks and leak detection for all enclosures where mechanical connections and single contained tubing are present. Door interlock and leak detection shall immediately close automated chemical supply valves, initiate a local alarm, shut down supply and de-pressurize system. If systems are designed with isolated primary and secondary engines for the purpose of redundancy with seperate engine leak detection, the affected engine shall de-pressurize and shutdown without impacting the redundant engine. 2. Removable spray/splash guards shall be used to shield technicians from pressurized components during maintenance. 3. Dispense/Blend units shall be equipped with sampling devices designed to minimize contamination and exposure during sample collection. <u>Day Tank:</u> 1. Spray or splash guards shall be installed on any connections on the top or bottom of the day tanks. The guards can be heavy vinyl curtains, white/clear PVC barriers, or flange/valve safety spray shields. All transfer/recirculation/inter-connect PFA(primary) lines shall not be exposed and shall have unbroken continuous double containment up to all the connection points within the spray/splash guards. 2. External sight tubes shall also require spray/splash protection. The preferred method is to have a clear PVC double containment for the sight tubes(schedule 40 or greater). <u>Drum/Tote Stations:</u> 1. Spray/splash guards shall be installed around the drum or tote connections or shall have a shielding to enclose the drum/tote. The shielding can be heavy vinyl curtains or poly/clear PVC barriers around the drum/tote. 2. The drum/tote PFA(primary) transfer/recirculation lines shall not be exposed and shall have an unbroken continuous double containment up to all the connections points within the spray/splash guards.		
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2.4.6 Installation (Cont'd)	<u>Distribution System:</u> 1. Distribution systems shall be individually contained and not allowed to be routed in chemical trays. Existing distribution, PFA(primary) lines shall not be exposed and shall be re-routed with individual double containment in a continuous unbroken clear PVC secondary containment (schedule 40 or greater) from the CDU to each valve manifold box. 2. Valve manifold boxes (VMB) doors must be secured with mechanical latches. 3. Valve manifold boxes (VMB) shall have leak detection in place. a) If VMB is designed with a single leak sensor, the CDU shall de-pressurize and stop supply upon leak activation. The VMB shall have appropriate manual isolation valves in place to safely replace the leaking valves or rectify leaking fittings with PPE. b) If VMB is designed with dual leak detection system, it shall be configured as low level(warning) and high level (shutdown). Upon activation of both low and high level leak sensors, the CDU shall de-pressurize and stop supply. The VMB shall have appropriate manual isolation valves in place to safely replace the leaking valves or rectify leaking fittings with PPE. c) If VMB is designed with dual leak detection system with an automatic VMB isolation valve, upon activation of low level leak sensor, the automatic VMB isolation valve shall close to stop supply to VMB. When both low and high level leak sensors are activated, the CDU shall de-pressurize and stop supply. The VMB shall have appropriate manual isolation valves in place to safely replace the leaking valves or rectify leaking fittings with PPE. 4. All instruments or other mechanical connections shall be installed with secondary containment and/or containment boxes.		
2.4.6 Installation (Cont'd)	<u>Labels:</u> TMAH labels and Hazard warning labels must clearly identify areas where TMAH or TMAH as a component of a mixture with concentrations >1% may be encountered in facilities, sub-fab and production areas. Sites, if required, may use both local and English language for TMAH Hazard warning labels. Both languages may be incorporated in the same label space or as a separate label, if constrained by size and the main requirement is to ensure that the label is clearly visible and legible from ground level. <u>A) Labels – Facilities Chemical System and Process Tools(located in Fab & Sub-fab) :</u> 1. TMAH labels shall clearly indicate both the <u>Chemical name/notation</u> and <u>TMAH % concentration</u> in orange label with black text (Example: <u>TMAH 25%</u> or <u>DEV4262 (TMAH 2.38%)</u>, as shown below).  Labeling convention, an example 2. The labels shall be clearly legible and visible and shall be pasted on all of the following systems: a) Bulk Storage/ Day tanks. b) Chemical Dispense Unit. c) Blend Unit. d) Drum/Tote Transfer Unit/Stations. e) Clean Coupling Booth(CCB). f) VMB, Tee/Junction box, End/Expansion-box for distribution and Containment boxes at top and bottom of storage/Blend tank/Day tank. g) Double contained supply/return piping and all other system inter-connect contained pipings including hook-up lines from VMB to tool chemical cabinet and/or process tool. h) Process Tool chemical cabinet and Main Process tools.		

2.4.6 Installation (Cont'd)	3. All double contained pipings shall be labeled with the chemical contained and the direction of flow. Two sizes of labels are available: for 1 inch secondary containment and containment that is greater than 1 inch. a) Piping = 1 inch, the lettering shall be bold, black, non-serif, 1 inch high (78 pt), Helvetica font on a 1.5 inch wide orange adhesive label. b) Piping >1 inch, the lettering shall be bold, black, non-serif, 1.5 inches high (114 pt), Helvetica font on a 2.5 inch wide orange adhesive label. 4. All containments shall be labeled at every 5 meters or shorter and on both sides of any penetrations or changes in direction. Labels shall be clearly visible and legible at ground level. 5. As a minimum guideline, all other systems indicated above(section A, s/n #2) shall be labeled with bold, black, non-serif, 1 inch high (78 pt), Helvetica font on a 1.5 inch wide orange adhesive label. 6. The orange hazard warning label to indicate the risk of TMAH exposure shall be placed on all equipment access panels/doors on all of the above mentioned systems(section A, s/n #2) with the exception of double contained pipings. The hazard warning label shall have the verbiage "WARNING – CONTAINS TMAH – AVOID SKIN CONTACT" in orange label and black text as shown below. The label shall be sized such that is clearly visible and legible from ground level and shall be placed in close proximity to equipment access panel/door's handle/knob where possible or in clear sight of the operator.  Orange Hazard Warning Label Black and Orange Tape (Zebra Tape)		
2.4.6 Installation (Cont'd)	7. Black and orange tape(zebra tape) as shown above shall be placed on the floor/tiles around the perimeter of Process tool chemical cabinets and Main Process tools or around the specific section of the system/tool that contains TMAH or TMAH as a component of a mixture with concentrations >1%. B) Labels – Waste Stream and Exhaust Ductwork/Condensate Drain 1. Hazard warning labels shall be placed on Exhaust ductwork at tool POC, Exhaust condensate drains and Waste streams to identify any exhaust ductwork and drain lines that has a "Potential" for TMAH concentrations >1%.  Hazard Warning Label for Waste and Exhaust systems 2. The hazard warning label indicating "Potential for TMAH concentrations >1%" shall be used for: - Process tool drain - Tool chemical cabinet/support equipment drain lines - Drain lines from Chemical dispense unit (CDU), Blend unit, Drum/Tote Transfer unit/station - Room sump pump and drain lines - Valve Manifold Box (VMB) drain line (if connected to submain) - Submain and hook-up drain lines - Final collection/storage, treatment tanks (if applicable to site) - Point of connection (POC) exhaust ductwork at Process Tool - Point of connection (POC) exhaust ductwork at Chemical Distribution System		

2.4.6 Installation (Cont'd)	3. For drain pipes/POC exhaust duct size that are greater than 3 inch and for collection tanks: The total dimension of the above label shall be 250mm by 100mm and it shall include the signal word for warning, identification of the hazard, and a pictogram to display the consequence. a) Signal word "WARNING" shall be in black lettering of 3/4 inch high, bold in Arial font on top right hand corner with an orange background of dimension 140mm x 35mm. b) Hazard Identification (spelling) "Potential for TMAH >1%" shall be in black lettering of 5/8 inch high, bold in Arial font at bottom right corner with a white background of dimension 140mm x 65mm. c) Pictogram indicating toxicity on the left with dimension of 110mm x 100mm. 4. For drain pipes/POC exhaust duct size below 3 inch: The total dimension of the above label shall be 125mm by 50mm and the fonts sized to match the label dimension. For smaller diameter drain pipes where the label size is too big, it shall be laminated with labels at both front and back and cable tied to the pipe. 5. All submain drain lines shall be labeled at no more than 20 ft and on both sides of any penetrations or changes in direction. Labels shall be clearly visible and legible at ground level. 6. Tool hook-up drain line shall be labeled at both tool POC and at the connection to submain (Y-Tee or Tee connection). 7. All hazard warning labels shall be clearly visible and legible from ground level.		
2.4.7 Other	<u>Others:</u> 1. System exhaust from TMAH CDU/tote/drum enclosures shall be connected to scrubbed caustic/ammonia exhaust. 2. All systems shall have leak detection. <u>Delivery:</u> 1. Drums and totes shall be delivered to the site with connections covered. 2. All totes/drums shall be delivered with dip tubes pre-installed. 3. Bottom drain valves of totes shall be protected during transport. If possible, it is preferred not to have a manual drain valve at the bottom of totes to eliminate the risk of leaks. 4. Totes shall be delivered in a protective cage where possible, to minimize puncture risk from forklifts. 5. Refer to Global EHS – TMAH Safety Standard for more details on safe operation during delivery <u>Storage:</u> 1. Separate TMAH from other incompatible chemicals by distance or physical separation as required by building/local codes. 2. Storage areas shall be well maintained. Adequate space shall be provided for the equipment used to transport chemicals safely. 3. Containers shall be tightly closed. TMAH readily absorbs carbon dioxide from the air. Loose caps present a spill risk. 4. TMAH shall be stored on the floor or bottom rack of shelving units. 5. If stored on higher shelves, the racking system shall be of an open grating style.		

2.5 Gas			
2.6 Water & Waste Water			
2.6.1 Maintenance	Maintenance, repair, and construction of drains shall include, but not be limited to, the following items: 1. Technicians performing maintenance, repair, and construction shall wear the Minimum Corporate Required PPE as stipulated in Global EHS – TMAH Safety Standard 2. Isolation or diversion valves shall be installed on the main drain line downstream of the last sub-main or lateral connection and prior to the at end of the drain lines above the collection tanks to allow for maintenance. 3. Maintenance, repair and construction on a drain lateral shall require isolation of chemical supply to the tools connected to the drain lateral. Chemical supply shall be isolated in accordance with site CoHE(Control of Hazardous Energy) procedures prior to work on a drain lateral. 4. An emergency recovery procedure shall be implemented to guarantee a safe and efficient response to a TMAH chemical leak/spill event. It shall be coordinated using the TMAH Emergency Response / Spill and TMAH Emergency Response/First Aid – Medical reference sheet as stipulated in Global EHS TMAH Safety Standard		
2.6.2 Materials	1. PVDF material is not suitable for TMAH waste lines. For more details, refer to material compatibility section 2.4.9.		
2.6.3 Installation	Labels: 1. All drain lines for Manufacturing tools or associated tool support equipment supplied by any combination of chemistries that include TMAH at concentrations above 1% shall be labeled with "Potential for TMAH >1%" hazard warning label and the direction of flow. 2. TMAH waste collection drains shall be identified with "Potential for TMAH >1%" hazard warning label and the direction of flow from the point of connection at the tool to the eventual point of final collection, storage, or treatment to eliminate the hazards related to this chemistry. However, if the TMAH concentration is deemed to be < 1% at the final Wastewater Treatment, no labeling is required. 3. The waste streams/systems shall be labeled but not limited to the following: a) Process tool drain lines. b) Tool chemical cabinet/support equipment drain lines c) Drain lines from Chemical dispense unit (CDU), Blend unit, Drum/Tote Transfer unit/station. d) Room sump pump and drain lines. e) Valve Manifold Box (VMB) drain line (if connected to submain). f) Submain and hook-up drain lines. c) TMAH waste collection tanks and associated lines (applicable to sites that has a dedicated waste TMAH collection and treatment system). 4. For more details of labeling guidelines, refer to section 2.4.12 under "Labels", sub-section B. Installation of Waste TMAH Lines: 1. a) Gravity drain lines carrying TMAH waste shall be hard piped connections to the facility drain system. The piping material shall be PVC schedule 80 or HI-PVC or PP. b) Pressurized drain/transfer lines that utilizes PFA tubing, PVC or PP to contain the waste stream, shall have either PE(for outdoor use) or clear PVC(for indoor use) as a double containment. The double containment shall be unbroken continuous containment up to all the connection points. c) All pressurized drain/transfer lines that are hard piped without double containment shall have a splash/spray guard around the flange connections.		

2.7 Other – Process Tools/ Labeling Examples			
2.7.1 Design Configuration – Process Tool	<u>All Process tools using TMAH or TMAH as a component of a mixture >1% concentration:</u> 1. All process tools shall protect and contain all process piping within the tool or chemical cabinet enclosure. 2. The tool and chemical cabinets shall be provided with door interlocks for hinged door with latch/handle and leak detection for all enclosures where mechanical connections and single contained tubing are present. Door interlock and leak detection shall immediately close automated chemical supply valves on the tool, initiate a local, alarm, notify operator/owner and shut down pneumatic supply to the tool chemical pumps and/or de-pressurize tool pressure pot systems.		
2.7.2 Maintenance – Process Tool	1. TMAH Process tools shall be maintained per manufacturer's recommendations. 2. Maintenance procedures shall include, but not be limited to the following items: 2.1 Technicians performing maintenance shall wear the Minimum Required PPE as stipulated in Global EHS – TMAH Safety Standard. 2.2 Areas where chemicals containing TMAH may be pressurized, it must be depressurized and LOTO applied prior to maintenance work. CoHE procedures for maintenance must be in place. 2.3 Means shall be available to verify all components have been depressurized before removing any protective cover. 2.4 Where necessary to read gauges or displays behind covers, covers shall be clear and adequately transparent to read the gauge or display without removing the cover. 2.5 Components that may need to be dismantled for maintenance, or repair, or replacement shall be flushed prior to being dismantled. 2.6 Components or fittings that need to be disconnected during maintenance shall be drained and will not result in liquid spilling onto maintenance personnel. 2.7 An emergency recovery procedure shall be implemented to guarantee a safe and efficient response to a TMAH chemical leak/spill event. It shall be coordinated using the TMAH Emergency Response/Spill and TMAH Emergency Response/First Aid – Medical reference sheet as stipulated in Global EHS –TMAH Safety Standard.		
2.7.3 Installation – Process Tool	1. All process pipings running to the process tools or chemical cabinets and tool inter-connects shall be installed with continuous secondary containment capable of directing liquid to a cabinet or enclosure with leak detection. 2. Drain lines carrying TMAH waste shall be hard piped and/or double contained to the facility drain system. Refer to sections 2.6.9 and 2.6.12 for more details. 3. Any drip pans installed under the tools shall have a hard piped and/or double contained connection to the facility drain system. 4. A leak detection sensor shall be installed in the drip pan. There shall be no mechanical fittings outside of containment. All connections through containment pans must be welded feed-through type connections. 5. Labels highlighting the presence of TMAH shall be pasted on process tools, chemical cabinets, all double contained inter-connect pipings and drain lines. Refer to section 2.4.12 for more details on labelling guidelines. 6. Secondary containment for tool hook-up lines and tool chemical cabinets in the sub-fab shall utilize schedule 40, clear PVC and/or shall utilize Poly Vinyl braided flexible type if constrained by space and installation difficulties. The secondary containment connections shall be secured by utilizing appropriate bulk head connections where possible or pipe clips. 7. Process tool exhaust ductwork and exhaust drain piping utilizing TMAH shall be labeled in accordance with section 2.4.12. Exhaust condensate drain piping does not require double containment or drip pan. 8. For more details of labeling guidelines, refer to section 2.4.12 under "Labels", sub-section A & B.		

2.7.4 Other –
Labeling Examples

Sample of acceptable TMAH Labeling for Facilities Systems & Process Tools across the Micron Sites.

TMAH Labels on Hook-up line

Supply line from VMB to Chemical cabinet with double containment & orange TMAH labeling.

Flexible braided double containment from Chemical cabinet to Photo tool.

2.7.4 Other –
Labeling Examples
(cont'd)

TMAH Labels

TMAH labels indicating the concentration shall be pasted on all :

- Supply piping
- TMAH storage tanks
- Dispense/Blending/Transfer systems
- VMBs
- Tool chemical cabinet
- Process tools

Hazard Warning labels shall be pasted on all:

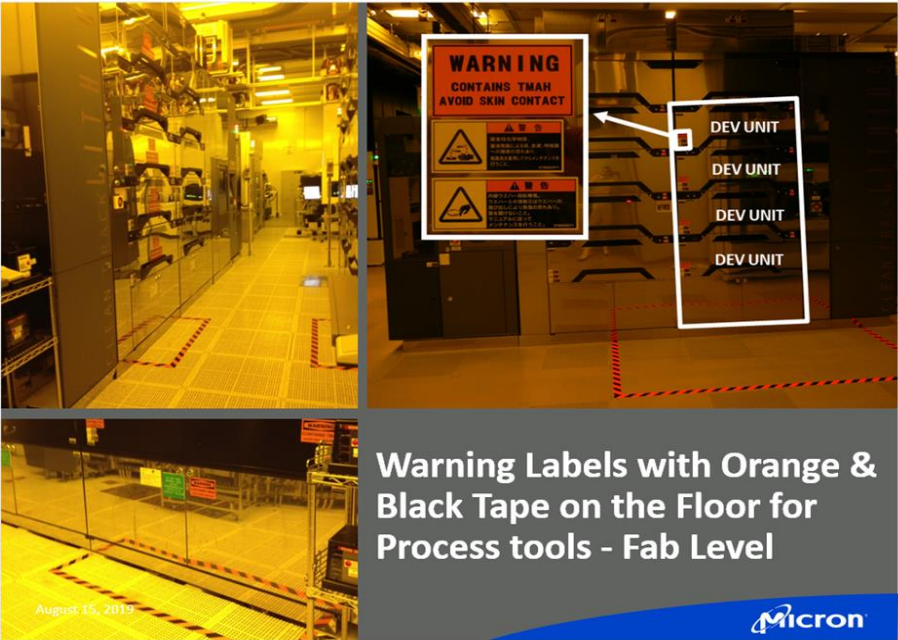
- TMAH storage tanks
- Dispense/Blending/Transfer systems
- VMBs
- Tool chemical cabinet
- Process tools

2.7.4 Other –
Labeling Examples
(cont'd)



TMAH Warning Labels installed on Day Tank & VMBs

2.7.4 Other –
Labeling Examples
(cont'd)



Warning Labels with Orange & Black Tape on the Floor for Process tools - Fab Level

August 15, 2019



3 Best Practices	Hide or Display Rows as Needed		
3.1 Mechanical			
3.2 Electrical			
3.3 I & C			
3.4 Chemical	Best Practices – applicable for new system installations if it does not exist in existing systems and over and above the minimum requirements already described.		
3.4.1 Operation	<u>Drum and Totes Connection:</u> 1. Quick disconnects shall be used for diptube connection. <u>Tanker Truck Unloading:</u> 1. Quick connect coupler connections shall be used for unloading to Clean Coupling Booths (CCBs). 2. Safety clamps shall be used to prevent the unloading hose with coupler from dislodging from the CCB (high pressurized hoses could endanger personnel if dislodged during high pressure unloading operations).		
3.4.2 Installation	<u>Drums/Totes Stations:</u> 1. Drums or totes of TMAH or TMAH as a component of a mixture with a concentration >1% will be installed in exhausted enclosures with leak detection. The enclosure shall have a built-in double glove access port with chemical resistant gloves attached. <u>TMAH Chemical Dispense/Blend Units:</u> 1. Chemical dispense/blend units shall be installed in chemical compatible, exhausted enclosures. 2. The enclosures shall have leak detection sensors to stop chemical supply upon activation. 3. The enclosures shall be compartmentalized to segregate between primary and secondary engines with a permanent spray/splash guards or a separation plate to shield technicians from pressurized components during maintenance. 4. Chemical dispense/blend units shall be equipped with sampling compartment with door interlocks designed to stop sample collection when sample door is opened. This is to minimize TMAH exposure during sample collection. <u>Day Tanks:</u> 1. A permanent enclosure with leak detection sensors shall be installed around the top and bottom connections of the day tanks. <u>Distribution Systems:</u> 1. For all new installations, TMAH distribution PFA piping shall be double contained in a continuous PVC secondary containment (schedule 40 or greater) and all POC valves, instruments, or other mechanical connections are to be enclosed in a valve manifold box (VMB) or Tee-box with leak detection to stop the chemical supply upon activation. 2. Each POC stick from the main manifold inside the VMB shall comprise of a manual valve(MV) followed by a sample/purge tee-off valve and finally an air operated valve(AOV). The AOV shall operate based on tool demand. 3. VMB access cover panels shall have keyed locks in place to prevent unauthorized access.		